11 (a) Expand and simplify $(x + 2)(3x - 1)(x + 4)$

	$3x$	-1
x	$3x^2$	$-x$
2	$6x$	-2

1 mark
Multiply every grid cell.

$(x + 2)(3x - 1) = 3x^2 + 5x - 2$

$(3x^2 + 5x - 2)(x + 4)$

1 mark
Multiply all terms again.

$= 3x^3 + 5x^2 - 2x + 12x^2 + 20x - 8$

$= 3x^3 + 17x^2 + 18x - 8$

1 mark
Collect like terms.

.....

(3)

(b) Solve $\frac{23 - 2y}{4} = y - 7$

$23 - 2y = 4(y - 7)$

1 mark
Multiply both sides by 4.

$23 - 2y = 4y - 28$

$51 = 6y$

1 mark
Collect y terms.

$y = 8.5$

1 mark
8.5.

$y =$ .....

(3)

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(c) Solve $3p^2 - 3p - 7 = 0$

Give your solutions correct to 3 significant figures.

$a = 3, b = -3, c = -7$

1 mark
Substitute with signs.

$$p = \frac{-(-3) \pm \sqrt{((-3)^2 - 4 \times 3 \times (-7))}}{2 \times 3}$$

$$= \frac{3 \pm \sqrt{93}}{6}$$

1 mark
Discriminant 93.

$p = 2.107275126...$ or $p = -1.107275126...$

$p = 2.11$ or -1.11 (3 significant figures)

1 mark
Both roots rounded.

(3)

(Total for Question 11 is 9 marks)

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